

Radical Equations and Extraneous Roots

Isolating and undoing a radical, testing every candidate in the original equation, and solving radical models from real measurements

Directions

- Isolate the radical first, then raise both sides to the matching power.
- Squaring is not reversible: it can produce candidates the original equation never had. **Every** candidate must be tested in the original equation before it is reported as a solution.
- A square root is never negative. If the isolated radical is set equal to a negative quantity, that candidate cannot work.
- In the modelling problems the units are given. Give each answer in the unit asked for, rounded as stated.

1. Solve for x : $\sqrt{x+5} = 4$

Answer: _____

2. A stone is dropped from a bridge. The fall time t , in seconds, is modelled by $t = 0.45\sqrt{d}$, where d is the drop height in metres. The stone takes 2.7 seconds to reach the water. Find the drop height.

Answer: _____ m

3. Consider the equation $\sqrt{2x+3} = x$.

(a) Solve it, reporting only the values that satisfy the original equation: _____

(b) Squaring also produces the candidate $x = -1$. Evaluate the left side, $\sqrt{2x+3}$, at $x = -1$: _____

4. Solve for x : $2\sqrt{x-1} + 3 = 11$

Answer: _____

5. Accident investigators estimate a car's speed s , in miles per hour, from the length d of its skid marks, in feet, using $s = \sqrt{21d}$. A car left skid marks consistent with a speed of 42 miles per hour. Find the length of the skid marks.

Answer: _____ ft

6. Consider the equation $\sqrt{x+7} = x-5$.

(a) Solve it, reporting only the values that satisfy the original equation: _____

(b) Squaring also produces the candidate $x = 2$. Evaluate the left side, $\sqrt{x+7}$, at $x = 2$: _____

(c) Explain why the candidate $x = 2$ has to be rejected. Refer to the value of the right side there.

7. Solve for x : $\sqrt{3x+1} = \sqrt{x+9}$

Answer: _____

8. From a height of h metres above sea level, the distance D to the horizon, in kilometres, is modelled by $D = 3.57\sqrt{h}$.

(a) A coastguard lookout must be able to see 35.7 kilometres. Find the height the lookout must be built at.

Answer: _____ m

(b) A second lookout stands 250 metres above sea level. Find its sighting distance, rounded to two decimal places.

Answer: _____ km

9. Consider the equation $\sqrt{x+4} + 2 = x$.

(a) Solve it, reporting only the values that satisfy the original equation: _____

(b) Squaring also produces the candidate $x = 0$. Evaluate the left side, $\sqrt{x+4} + 2$, at $x = 0$: _____

10. Challenge. Consider the equation $\sqrt{x+5} - \sqrt{x} = 1$.

(a) Solve it. (Isolate one radical, square, then isolate the remaining radical and square again.): _____

(b) Evaluate $\sqrt{x+5} - \sqrt{x}$ at your solution to confirm it satisfies the original equation: _____